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Predictive Modeling and Risk Scoring for Bank Customer Churn

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**A Machine Learning-Based Churn Intelligence System for Retail
Banking Customer Retention**

Abstract

Customer churn is a major challenge for retail banks because it directly affects customer lifetime value, revenue stability, cross-selling opportunities, and long-term competitiveness. Traditional churn analysis usually focuses on explaining why customers have already left, but modern banking requires proactive systems that can identify customers at risk before churn occurs.

This project develops a predictive churn intelligence system using machine learning. The system predicts whether a bank customer is likely to churn, generates a churn probability score, assigns a risk category, and provides actionable retention recommendations. The dataset contains 10,000 European bank customer records with demographic, financial, product usage, and engagement-related attributes.

Abstract

The methodology includes data cleaning, feature engineering, exploratory data analysis, stratified train-validation-test splitting, preprocessing, hyperparameter tuning, threshold optimization, model comparison, probability calibration, and Streamlit dashboard deployment. Multiple models were trained and compared, including Logistic Regression, Decision Tree, Random Forest, Extra Trees, Gradient Boosting, Hist Gradient Boosting, and XGBoost.

After tuning, the best raw model was Tuned Gradient Boosting, achieving a test ROC-AUC of 0.8682. For final deployment, the Calibrated Tuned Gradient Boosting model was selected because the project requires reliable probability-based churn risk scoring. The calibrated model achieved an accuracy of 0.8400, precision of 0.5960, recall of 0.6634, F1-score of 0.6279, ROC-AUC of 0.8667, and PR-AUC of 0.7095.

Abstract

The project concludes that customer churn is strongly influenced by age, number of products, engagement-product interaction, active membership status, geography, balance, age-tenure interaction, and balance-to-salary ratio. The final Streamlit dashboard enables customer-level churn risk prediction, what-if scenario simulation, feature importance visualization, and model performance comparison.



Introduction

Customer churn refers to the loss of customers who discontinue their relationship with a business or service provider. In retail banking, customer churn is particularly important because acquiring a new customer is usually more expensive than retaining an existing one. Churn also reduces customer lifetime value, weakens revenue consistency, and limits opportunities for cross-selling and upselling financial products.

Banks collect large amounts of customer-level data, including demographic details, account balances, credit scores, product usage, activity status, and salary estimates. However, many banks still rely on reactive retention strategies, where intervention happens only after a customer has already shown strong signs of disengagement or has exited the bank.

This project aims to transform churn analysis from a reactive process into a predictive risk intelligence system. Instead of simply classifying customers as churned or retained, the system estimates churn probability and converts it into a practical risk category. This allows banks to prioritize customers based on risk level and design targeted retention campaigns.

Problem Statement

Despite having rich customer data, banks often lack accurate and explainable

systems for identifying customers who are

- ☐ likely to churn. Existing
- ☐ churn analysis
- ☐ approaches may
- ☐ describe historical
- ☐ churn patterns, but
- ☐ they do not always
- ☐ provide real-time or
- ☐ forward-looking insights.

PROB 1

Lack of accurate churn prediction models.

PROB 2

Absence of customer-level churn probability scores.

PROB 3

Limited interpretability of churn drivers.

PROB 4

Reactive retention strategies instead of proactive intervention.

PROB 5

Inefficient broad-based campaigns instead of targeted retention actions.

PROB 6

Difficulty in understanding how changes in engagement or product usage affect churn risk.

Objectives

1 Primary Objectives

The primary objectives of this project are:

1. To predict customer churn using machine learning models.
2. To generate churn probability scores for individual customers.
3. To identify key factors driving customer churn.
4. To build an interactive dashboard for churn risk analysis.

2 Secondary Objectives

The secondary objectives are:

1. To reduce false positives and improve model reliability.
2. To improve interpretability through feature importance analysis.
3. To support scenario-based churn risk analysis.
4. To create a risk-based customer retention framework.
5. To support business decision-making for retail banking customer relationship management.

Dataset Description

The dataset used in this project contains 10,000 European bank customer records. Each row represents a customer, and each column represents customer-level information related to demographics, account relationship, product ownership, banking activity, and churn status.

- 1) Year
- 2) customerID
- 3) Surname
- 4) Credit score
- 5) Geography
- 6) Gender
- 7) Age
- 8) Tenure
- 9) Balance
- 10) Num Of product
- 11) Has Cr Card
- 12) ActiveMember
- 13) Exited

Target Variable

The target variable is **Exited**.

0 Customer Retained

1 Customer Churned

The target distribution showed that 79.63% of customers were retained and 20.37% of customers churned. This indicates a class imbalance problem, where churned customers form the minority class.

Data Preprocessing



Data preprocessing was performed to prepare the dataset for machine learning models



1 Missing Value Check



The dataset had no missing values across all columns. Therefore, no major missing value imputation was required.



2 Duplicate Check



The dataset had no duplicate rows. Therefore, no duplicate removal was required.



3 Removal of Non-Informative Columns



The following columns were removed:



CustomerId



Surname



Year



After removing these columns, the dataset was reduced to predictive variables only.

Data Preprocessing

4 Categorical Encoding

The categorical variables were:

1. Geography
2. Gender

One-hot encoding was applied to convert categorical variables into numerical form. The final encoded features included:

1. Geography_Germany
2. Geography_Spain
3. Gender_Male

France and Female were treated as reference categories after encoding.

5 Feature Scaling

Numerical features were scaled using StandardScaler. Scaling was especially important for models such as Logistic Regression, where feature magnitude can influence coefficient estimation.

Feature Engineering

Feature engineering was performed to create additional business-relevant variables that capture customer relationship strength, product intensity, financial behavior, and engagement.

The engineered features were:

1 Balance-to-Salary Ratio

This feature measures the relationship between a customer's account balance and estimated salary.

A higher ratio may indicate stronger financial linkage with the bank, while a lower ratio may indicate limited account engagement.

2 Product Density

This feature measures the number of products used per year of tenure.

It captures how deeply the customer has adopted the bank's products relative to the length of the relationship.

3 Engagement-Product Interaction

This feature combines active membership status with number of products.

It captures whether the customer is both active and product-engaged.

Feature Engineering

4 Age-Tenure Interaction

This feature combines customer age with tenure.

It helps capture relationship maturity across different age groups.

5 High Balance Flag

This binary feature indicates whether a customer has a balance above the median balance.

6 Zero Balance Flag

This binary feature indicates whether a customer has zero account balance.

7 Senior Customer Flag

This feature identifies customers aged 60 and above.

8 Low Credit Score Flag

This feature identifies customers whose credit score is below the median credit score.

After feature engineering, the dataset contained 19 columns, including the target variable.

Exploratory Data Analysis

Exploratory Data Analysis was performed to understand churn behavior across customer segments.

1 Churn Distribution

The dataset showed a churn rate of 20.37%. This confirms that the dataset is imbalanced, with retained customers forming the majority class.

This imbalance makes accuracy alone insufficient as an evaluation metric. A model can achieve high accuracy by predicting most customers as retained, but such a model may fail to detect actual churners.

2 Churn by Geography

Churn varied across countries. Germany showed the highest churn rate among the three geographies. This indicates that customer behavior and market conditions may differ by geography, making location an important factor in churn prediction.

3 Churn by Gender

Female customers showed a higher churn rate compared to male customers. However, gender alone should not be interpreted as the primary cause of churn. It should be considered along with engagement, product usage, geography, and financial behavior.

Exploratory Data Analysis

	Model	Best F1 Threshold	Best Business Threshold	Best F1	Best Business Score
0	Tuned Logistic Regression	0.60	0.11	0.509642	0.593878
1	Tuned Decision Tree	0.19	0.18	0.579677	0.637464
2	Tuned Random Forest	0.34	0.17	0.630872	0.648657
3	Tuned Extra Trees	0.30	0.16	0.622419	0.650307
4	Tuned Gradient Boosting	0.33	0.17	0.629870	0.647551
5	Tuned Hist Gradient Boosting	0.31	0.13	0.621795	0.634656
6	Tuned XGBoost	0.65	0.37	0.638574	0.648732

4 Churn by Active Membership

Inactive members had a higher churn rate compared to active members. This supports the idea that engagement is one of the strongest indicators of customer retention.

5 Churn by Number of Products

Customers with 3 or 4 products showed significantly high churn risk. This is an important business insight because it suggests that product quantity alone does not always imply loyalty. In some cases, customers with multiple products may face dissatisfaction, complexity, or unsuitable product bundling.

6 Age and Churn

Age showed a strong relationship with churn. Older customers were more likely to churn compared to younger customers. This made Age one of the most important predictors in the final model.

7 Balance and Churn

Balance also showed a relationship with churn. Customers with higher balances may represent valuable customers, but they may also be more sensitive to service quality, pricing, or competing offers from other banks.

Train-Validation-Test Strategy

The dataset was split using stratified sampling to preserve the churn distribution across training, validation, and testing sets.

The splitting strategy was:

1. Training set: used for model fitting and cross-validation.
2. Validation set: used for threshold optimization.
3. Test set: used for final unbiased model evaluation.

Stratification was important because the churn class represented only about 20% of the data. Without stratification, some splits could contain a distorted churn ratio and produce unreliable evaluation results.



Model Development

1 Models Used

The models used in this project were:

1. Logistic Regression
2. Decision Tree
3. Random Forest
4. Extra Trees
5. Gradient Boosting
6. Hist Gradient Boosting
7. XGBoost

2 Baseline Model

Logistic Regression was used as an interpretable baseline model. It provides coefficient-based explanations and helps establish a benchmark for more complex models.

3 Tree-Based Models

Decision Tree, Random Forest, and Extra Trees were used to capture non-linear relationships and interactions between customer features.

4 Boosting Models

Gradient Boosting, Hist Gradient Boosting, and XGBoost were used as advanced models. These models are powerful for structured tabular data and can capture complex relationships between features

Hyperparameter Tuning

To improve model performance, RandomizedSearchCV was used for hyperparameter tuning. Stratified 5-fold cross-validation was applied during tuning to maintain the churn distribution in every fold.

1 Tuning Strategy

The tuning process optimized hyperparameters such as:

1. Number of estimators
2. Learning rate
3. Maximum tree depth
4. Minimum samples per split
5. Minimum samples per leaf
6. Feature sampling strategy
7. Subsample ratio
8. Regularization parameters for XGBoost
9. Class weight settings where applicable

2 Scoring Metrics During Tuning

The following metrics were tracked:

1. ROC-AUC
2. PR-AUC
3. Precision
4. Recall
5. F1-score

ROC-AUC was used as the main refit criterion because it measures the ability of the model to distinguish churned customers from retained customers across thresholds.

Model Evaluation Metrics

Because the dataset is imbalanced, multiple evaluation metrics were used.

1 Accuracy

Accuracy measures overall correctness. However, it can be misleading when one class dominates the dataset.

2 Precision

Precision measures how many customers predicted as churners were actually churners. High precision reduces unnecessary retention actions on customers who are not likely to churn.

3 Recall

Recall measures how many actual churners were correctly identified. In banking churn prediction, recall is very important because missing a real churner may lead to customer loss.

4 F1-Score

F1-score balances precision and recall. It is useful when both false positives and false negatives matter.

Model Evaluation Metrics

5 ROC-AUC

ROC-AUC measures the model's ability to separate churned and retained customers across different thresholds.

6 PR-AUC

PR-AUC is especially useful for imbalanced datasets because it focuses on performance for the minority churn class.

	Model	Best F1 Threshold	Best Business Threshold	Best F1	Best Business Score
0	Tuned Logistic Regression	0.60	0.11	0.509642	0.593878
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Threshold Optimization And Final Model Comparison

The default classification threshold of 0.50 is not always optimal for churn prediction. A lower threshold may capture more churners, while a higher threshold may reduce false positives.

Threshold optimization was performed using the validation set. The best threshold was selected based on F1-score and business relevance.

For the Tuned Gradient Boosting model, the optimized threshold was 0.33. This threshold provided a better balance between precision and recall than the default 0.50 threshold.

The final test results showed that Tuned Gradient Boosting achieved the highest ROC-AUC among the tuned models.

Threshold Optimization And Final Model Comparison

. The best raw model was Tuned Gradient Boosting.

Its final test performance was:

Metric	Value
Threshold	0.33
Accuracy	0.8525
Precision	0.6451
Recall	0.6118
F1 Score	0.6280
ROC-AUC	0.8682
PR-AUC	0.7113

Tuned Gradient Boosting was selected as the best raw model because it achieved the highest ROC-AUC, indicating the strongest class separation ability.

Probability Calibration and Final Model Selection

Since the project focuses on churn risk scoring, probability quality is important. The model should not only classify customers as churned or retained, but also provide a reliable churn probability score.

After model comparison, Tuned Gradient Boosting achieved the best raw performance with the highest ROC-AUC of 0.8682. However, for dashboard deployment and customer-level risk scoring, the calibrated version of this model was selected.

The final deployment model was:

Calibrated Tuned Gradient Boosting

Although calibration slightly reduced accuracy and precision, it improved recall. This is useful in churn prediction because identifying more actual churners is more valuable than simply maximizing overall accuracy.

Metric	Value
Threshold	0.30
Accuracy	0.8400
Precision	0.5960
Recall	0.6634
F1 Score	0.6279
ROC-AUC	0.8667
PR-AUC	0.7095

Feature Importance and Churn Drivers

Feature importance analysis was used to understand the major factors influencing churn prediction.

Rank	Feature	Importance
1	Age	0.318838
2	<u>NumOfProducts</u>	0.261072
3	<u>Engagement_Product</u>	0.091346
4	<u>IsActiveMember</u>	0.063453
5	<u>Geography_Germany</u>	0.048965
6	Balance	0.042263
7	<u>Age_Tenure</u>	0.036429
8	<u>Balance_to_Salary</u>	0.027350

The results show that churn is mainly influenced by customer age, product usage, engagement level, geography, balance, and customer-bank relationship strength. Age and number of products were the two most important predictors.

This suggests that churn is not only a demographic issue. It is also strongly connected with customer engagement, product relationship, and financial behavior.

Risk Scoring Framework

The final model generates a churn probability between 0 and 1. This probability was converted into business-friendly risk categories.

Churn Probability	Risk Category
0.00 – 0.30	Low Risk
0.30 – 0.60	Medium Risk
0.60 – 0.80	High Risk
0.80 – 1.00	Very High Risk

Risk Category	Recommended Action
Low Risk	Maintain regular engagement and standard service
Medium Risk	Send personalized engagement message or product recommendation
High Risk	Offer retention benefit, relationship-manager call, or product bundle
Very High Risk	Immediate retention intervention with personalized offer and priority support

Conclusion

This project developed a machine learning-based churn risk scoring system for retail banking customers. The goal was not only to predict whether a customer may churn, but also to generate a probability-based risk score and convert it into actionable business recommendations.

Multiple models were trained and tuned using stratified cross-validation and RandomizedSearchCV. Tuned Gradient Boosting achieved the best raw performance with a test ROC-AUC of 0.8682. For deployment, the Calibrated Tuned Gradient Boosting model was selected because it provides more reliable churn probability scores.

The final calibrated model achieved an accuracy of 0.8400, precision of 0.5960, recall of 0.6634, F1-score of 0.6279, ROC-AUC of 0.8667, and PR-AUC of 0.7095.

The most important churn drivers were Age, Number of Products, Engagement-Product interaction, Active Membership status, Geography Germany, Balance, Age-Tenure interaction, and Balance-to-Salary ratio.

Overall, this project reframes customer churn prediction as a proactive risk intelligence system. It helps banks identify high-risk customers early, understand churn drivers, and take targeted retention actions before customer loss occurs.

Business Insights and Recommendations

The analysis produced the following major business insights:

1. **Customer engagement is important**

Inactive customers showed higher churn risk. Banks should monitor activity levels and design engagement campaigns for inactive customers.

2. **Product usage needs careful interpretation**

Number of products was one of the strongest churn drivers. Customers with multiple products may still churn if product experience or bundling is poor.

3. **Older customers require attention**

Age was the most important churn driver. Banks should design age-sensitive retention strategies.

4. **Geography affects churn behavior**

Germany showed higher churn tendency. This suggests that banks may need region-specific retention policies.

5. **Balance behavior matters**

Balance and balance-to-salary ratio were important churn indicators. High-value customers should be monitored carefully because their churn can affect revenue stability.

Recommended actions include targeted retention campaigns, relationship manager follow-ups, personalized product offers, improved engagement programs, and special attention to high-risk customers.

Limitations and Future Scope

This project has some limitations. The dataset is static and does not include time-based customer behavior. Transaction history, complaint records, customer satisfaction data, and macroeconomic indicators were also not available.

In the future, this project can be improved by adding transaction-level features, customer complaint history, digital banking activity, SHAP-based individual explanations, customer lifetime value analysis, and automated monthly churn monitoring.

The dashboard can also be extended into a full CRM-style tool where banks can track high-risk customers and plan retention actions regularly.



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